

The support-recovery bottleneck

A common global model, decoder controls, and explicit ODMA gap

Stage	2 of 4
Updated	30 August 2026
Status	Jobs 001–017 complete; conclusions limited to tested algorithms and regimes

This report is a chronological research record. Measured results, interpretations, and open hypotheses are labelled separately. Detailed numerical ledgers remain in **results/**.

1 Why a common global model was needed

The first-stage decoder comparison mixed codebook structure and inference approximations. This stage held the recovery target fixed: every construction was represented by a unit-column matrix $\Phi \in \mathbb{C}^{n \times M}$ and the decoder estimated the same integer count vector $a \in \mathbb{Z}_+^M$ from

$$y = \Phi a + z, \quad \mathbf{1}^T a = K_a, \quad M = 2^B.$$

Dense and ODMA codebooks could therefore be compared through identical metrics and decoder interfaces.

2 The ideal and practical recovery problems

With known noise variance and load, exact maximum likelihood is

$$\hat{a}_{\text{ML}} = \arg \min_{a \in \mathbb{Z}_+^M, \mathbf{1}^T a = K_a} \|y - \Phi a\|_2^2.$$

The number of count multisets is $\binom{M+K_a-1}{K_a}$, so exact enumeration is only a tiny-system certificate.

The principal scalable baseline was nonnegative OMP with oracle K_a (NNOMP-OracleK). At iteration t , it selects a column using the residual correlation,

$$j_t = \arg \max_{j \notin S_t} \Re\{\phi_j^H r_t\},$$

then solves nonnegative least squares on S_{t+1} and updates $r_{t+1} = y - \Phi_{S_{t+1}} \hat{a}_{S_{t+1}}$. Its advantages are a transparent support-search mechanism and a matched interface across codebooks. Its oracle load and greedy selection mean that it is a practical baseline, not Bayes-optimal URA decoding.

Other explored routes—MAP-ADMM, block coordinate descent, AMP/VAMP variants, residual MAP, SIC, and BlockMAP—can be understood as different approximations to the same constrained objective. They were useful diagnostics, but none changed the central causal experiment: remove support search and measure what remains.

3 Metric and collisions

For true and estimated count vectors with equal total mass K_a , multiplicity-aware PUPE is

$$\text{PUPE}(a, \hat{a}) = 1 - \frac{1}{K_a} \sum_{m=0}^{M-1} \min(a_m, \hat{a}_m).$$

This reduces to missed-set fraction when every count is zero or one. At small B , repeated messages are not negligible, so a support-only metric would incorrectly penalise or ignore multiplicity. The project therefore distinguishes:

- a canonical count-aware experiment, where repeated messages are legitimate;
- low-collision large-alphabet comparisons, where set recovery is a close approximation;
- oracle-support tests, where the active indices are given but their counts still need estimation.

Bound overlays and empirical decoders answer different questions. A Polyanskiy-style achievability curve is a coding benchmark under its own collision convention; it is not an expected NNOMP curve and must not be used to certify an implementation merely because the numbers are close.

4 Explicit dense versus ODMA experiments

Jobs 001–008 and 013–014 used 50 seeds, NNOMP-OracleK, target PUPE ≤ 0.05 , and matched (B, n, K_a) for dense and ODMA codebooks. The gap was small in easy regimes and grew sharply as support recovery became compressed.

Geometry and load	Dense	ODMA 1	ODMA 2	ODMA 3
$B = 12, n = 2048, K_a = 333$	1.00	3.69	3.25	> 4
$B = 12, n = 512, K_a = 52$	2.19	11.56	6.44	unbracketed
$B = 12, n = 256, K_a = 17$	2.19	9.12	5.75	5.56

Table 1: Required E_b/N_0 in dB for PUPE ≤ 0.05 . ODMA arrangements use different local lengths/placement counts; the purpose is to show the practical gap, not to rank a universal ODMA parameter.

Increasing B at fixed n reduced repeated-message ambiguity while preserving the dense-ODMA separation. Reducing n at $B = 12$ amplified the gap. Thus the effect was not explained by the collision-heavy behaviour of very small alphabets.

5 Oracle-support decomposition

The decisive control supplies the active support $S = \{m : a_m > 0\}$ and solves

$$\hat{a}_S = \arg \min_{x \geq 0} \|y - \Phi_S x\|_2^2,$$

followed by integer projection. This removes the combinatorial search while retaining active-column conditioning, interference, noise, and multiplicity fitting.

Geometry	Mean oracle ODMA-minus-dense gap	Contrast with practical decoder
$B = 12, n = 2048$	0.05–0.11 dB	Practical gap reaches 2+ dB at high load
$B = 12, n = 512$	0.10, 0.12, 0.28 dB	NNOMP support loss is about 5–6 dB
$B = 12, n = 256$	0.08, −0.01, 0.03 dB	NNOMP support loss is about 6–7 dB

Table 2: Jobs 012, 016, and 017. Each range/list spans the tested ODMA arrangements.

Evidence status: Measured result. The true support is an explicit oracle; this is not an achievable receiver.

The active ODMA columns were not substantially worse than active dense columns once their identities were known. The large observed penalty under NNOMP therefore arose mainly during support selection. This conclusion is robust across the tested n , but it is neither an information-theoretic converse nor proof that every ODMA-aware decoder must suffer the same gap.

6 What this changed

The experiment redirected the project from adding ODMA message updates to asking which codebook structures are easy to search. The resulting requirements were: retain many distinguishable global supports; distribute row energy and overlap rather than reuse a tiny mask bank; and expose forward/adjoint operations without storing a dense $n \times 2^B$ matrix. The next stage introduced one factorised algebra for dense, ODMA, product, and sectioned constructions, with exact gradients through its continuous factors.